

CE 522: Matrix Methods for Structural Analysis

Course Code	CE 522
Course Name	Matrix Methods for Structural Analysis
L-T-P-C	3-0-0-3
Prerequisites	CE 404: Analysis of Structures/Equivalent
Intended for	UG Program (B. Tech 3rd;4th Year Students) and PG Program
Approval	52nd BoA (Nov 2023)

Course Contents

- **Module 1:** Introduction to matrix analysis of structures, elements and structures, degrees of freedom, the principle of superposition, concepts of matrix algebra. (3 Lectures)
- **Matrix analysis of structures with axial elements:** plane trusses and space trusses Flexibility method: Introduction to the approach, Deriving the flexibility matrix of truss. Analysis of determinate and indeterminate structures employing the flexibility matrix approach. (5 Lectures)
- **Matrix analysis of beams and grids:** Flexibility method-Deriving the flexibility matrix beams and grids: Analysis of determinate and indeterminate structures employing the flexibility matrix approach. (5 Lectures)
- **Matrix analysis of plane and space frames:** Flexibility method: Deriving the flexibility matrix of plane and space frames: Analysis of determinate and indeterminate structures employing the flexibility matrix approach. Analysis of prismatic and non-prismatic frames. (5 Lectures)
- **Matrix analysis of structures with axial elements:** plane trusses and space trusses. Stiffness Method: Introduction to the approach, Deriving the stiffness matrix truss. Analysis of determinate and indeterminate structures employing the stiffness matrix approach. (5 Lectures)
- **Matrix analysis of beams and grids: Stiffness method:** Derivation of stiffness matrix beams and grids: Analysis of determinate and indeterminate structures employing the stiffness matrix approach. (5 Lectures)
- **Matrix analysis of plane and space frames: Stiffness method:** Derivation of the stiffness matrix plane and space frames: Analysis of determinate and indeterminate structures employing the stiffness matrix approach. Analysis of prismatic and nonprismatic frames. (5 Lectures)
- **Module 8:** Comparison between stiffness and flexibility methods, Analysis of truss, beams, and frame structures using direct stiffness approach; Computer application of direct stiffness method. (5 Lectures)
- **Beyond matrix methods:** Introduction to finite element method, element types, basic formulation, and application to 1 D problems. (4 Lectures)

Text Books

1. Weaver, W., and Gere, J. M ., **Matrix analysis framed structures**, Springer science & business media, 2012.
2. Hibbeler, RC, **Structural Analysis**, 9th edition, Pearson Education, 2017.

References

1. G. S. Pandit and S. P. Gupta, **Structural Analysis: A Matrix Approach**, Tata McGrawHill, 2008.
2. Martin, H. C., **Introduction to Matrix Methods of Structural Analysis**, McGraw-Hill, New York, 1996
3. Menon, D, **Advanced Structural Analysis**, Narosa Publishing House, 2015.